

WTWIS8 Exam 2022:

Q1. $f(x) = \sqrt{x-2} - \sqrt{3-x}$

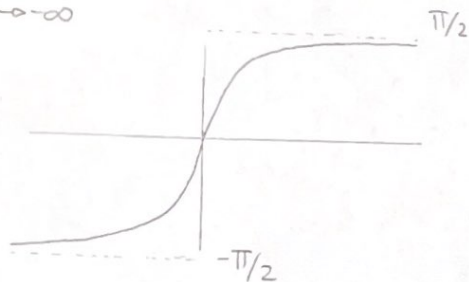
$$x-2 \geq 0 \quad \text{and} \quad 3-x \geq 0$$

$$x \geq 2 \quad x \leq 3$$

$$\therefore [2, 3]$$

f

Q2. $\lim_{x \rightarrow -\infty} \tan^{-1}(4x)$



$$-\pi/2$$

b

Q3. $f(x) = \ln(x) - x$

$$f'(x) = \frac{1}{x} - 1$$

for Critical numbers $f' = 0$ ① & f' DNE ②

① $\frac{1}{x} - 1 = 0$

$$\frac{1}{x} = 1$$

$$x = 1$$

② @ $x = 0$

but $x = 0$ is not in the domain of f .

$$c = 1$$

a

Q4. $y = x^3 + 4$

$$x = -1$$

$$y'(x) = 3x^2$$

$$y = 3$$

$$y'(-1) = 3$$

$$y = 3x + c$$

$$3 = 3(-1) + c$$

$$6 = c$$

$$\therefore y = 3x + 6$$

b

- Q5:
1. incorrect. as $\ln(ab)^2 = 2(\ln(a) + \ln(b))$
 2. correct as $\ln(6) - \ln(2) = \ln(6/2) = \ln(3)$
 3. incorrect as $\ln(e^x) = x \cdot \ln(e) = x$ and not 1

b

- Q6:
1. correct
 2. correct $f'(x) = 3\cosh(3x) + 3\sinh(3x) = 3(\cosh(3x) + \sinh(3x))$
 3. correct $f'(x) = 3e^x = 3f(x)$
 $f'(-x) = 3e^{-x} = 3f(-x)$

g

Q7: $\lim_{x \rightarrow \infty} x^2 e^{-x} \quad (\infty \times 0)$

$\lim_{x \rightarrow \infty} \frac{x^2}{e^x} \quad (\infty / \infty)$

$\lim_{x \rightarrow \infty} \frac{2x}{e^x} \quad (\infty / \infty)$

$\lim_{x \rightarrow \infty} \frac{x}{e^x} \quad (\infty / \infty)$

$\lim_{x \rightarrow \infty} \frac{1}{e^x}$

$= 0$

a

- Q8:
1. incorrect ; Domain is $\mathbb{R}$
 2. incorrect ; Domain is $(-\infty, 1)$
 3. correct

c

$$\int \frac{x^2}{x^2+1} dx$$

$$\int \frac{x^2+1-1}{x^2+1} dx$$

$$\int \frac{x^2+1}{x^2+1} dx - \int \frac{1}{x^2+1} dx$$

$$x - \arctan(x) + C$$

b

$$A(1, 2, 3)$$

$$B(3, 6, 7)$$

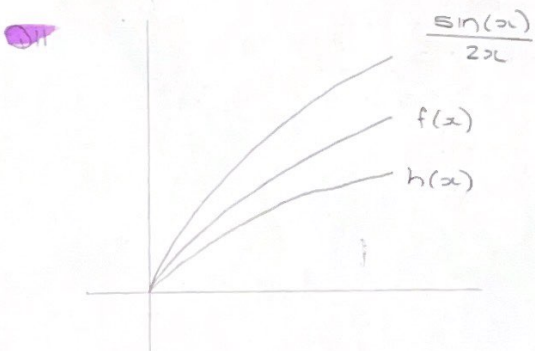
$$\vec{v} = \overline{AB} = \langle 3-1, 6-2, 7-3 \rangle$$

$$\vec{v} = \overline{AB} = \langle 2, 4, 4 \rangle$$

$$|\vec{v}| = |\overline{AB}| = \sqrt{2^2 + 4^2 + 4^2}$$

$$= 6$$

a



$$\lim_{x \rightarrow 0} \frac{\sin(x)}{2x} \quad (0/0)$$

$$\lim_{x \rightarrow 0} = \frac{\cos(x)}{2}$$

$$= 1/2$$

1. incorrect

2. incorrect

$$3. \lim_{x \rightarrow 0} \frac{\sin(3x)}{x} \times \frac{3}{3}$$

$$3 \lim_{x \rightarrow 0} \frac{\sin(3x)}{3x}$$

$$= 3$$

- correct

c

Q2 $f(x) = \frac{8x^3 + 5x^2 + 2x - 3}{1 - 2x^3}$

* degrees same $\therefore$ there is a HA

$$\lim_{x \rightarrow \infty} \frac{x^3(8 + 5/x + 2/x^2 - 3/x^3)}{x^3(1/x^3 - 2)}$$

$$= -4$$

$\therefore a$

SECTION B [Fill in questions] - 14 marks

SECTIONS B AND C MUST BE SCANNED AND UPLOADED ON CLICKUP.
Please do not scan your rough working or any empty pages.

SURNAME:

INITIALS:

STUDENT NUMBER:

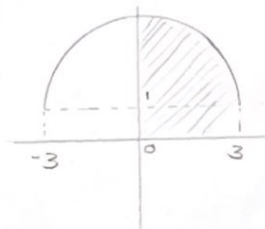
SECTION B: Write only your final answer in the block provided. Only this will be marked.

QUESTION 13 [2 marks]

Evaluate the integral by interpreting it in terms of areas.

$$\int_0^3 (1 + \sqrt{9 - x^2}) dx =$$

$$\frac{1}{4} \pi (3)^2 + 1(3) = \frac{9\pi + 12}{4}$$



QUESTION 14 [3 marks]

Find the derivative of the given function. Do not simplify your answer.

(a) $f(x) = e^x \ln x$

$$f'(x) = e^x \ln(x) + e^x \cdot (1/x)$$

(b) $f(x) = \cos(\cos x)$

$$f'(x) = \sin(\cos(x)) \cdot \sin(x)$$

(c) $f(x) = \frac{1}{\sqrt{x^2 + 1}}$

$$f(x) = (x^2 + 1)^{-1/2}$$

$$f'(x) = -\frac{x}{(x^2 + 1)^{3/2}}$$

QUESTION 15 [1 mark]

Find the value(s) of x such that the vectors $\langle 2x, 4, x \rangle$ and $\langle 3, 2, x \rangle$ are orthogonal.

$$\vec{a} \cdot \vec{b} = 0$$

$$\langle 2x, 4, x \rangle \cdot \langle 3, 2, x \rangle$$

$$6x + 8 + x^2 = 0$$

$$x^2 + 6x + 8 = 0$$

$$(x+2)(x+4) = 0$$

$$x = -2, x = -4$$

QUESTION 16 [4 marks]

Determine the following integrals:

(a) $\int \frac{\cos(\ln t)}{t} dt =$

$\sin(\ln(t)) + C$

(b) $\int \frac{x}{x^2+1} dx =$

$\frac{1}{2} \ln|x^2+1| + C$

$\int x(x^2+1)^{-1/2}$

(c) $\int \frac{x}{\sqrt{x^2+1}} dx =$

$\sqrt{x^2+1} + C$

(d) $\int 3^t \sec^2(3^t) dt =$

$\frac{1}{\ln(3)} \cdot \tan(3^t) + C$

QUESTION 17 [2 marks]

The equation of a sphere is:

$x^2 + y^2 + z^2 = 6x - 4y - 10z$

(a) The center of the sphere is:

(1)

$(3, -2, -5)$

(b) The radius of the sphere is:

(1)

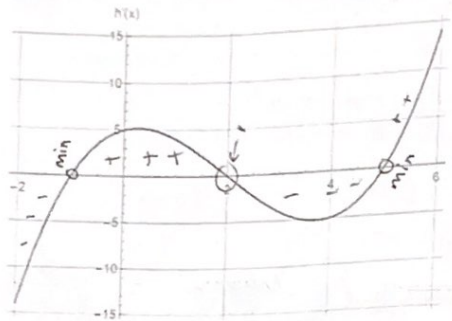
$r = \sqrt{38}$

$x^2 - 6x + y^2 + 4y + z^2 + 10z = 0 \quad \left(\frac{b}{2}\right)^2$

$(x^2 - 6x + 9) + (y^2 + 4y + 4) + (z^2 + 10z + 25) = 25 + 9 + 4$

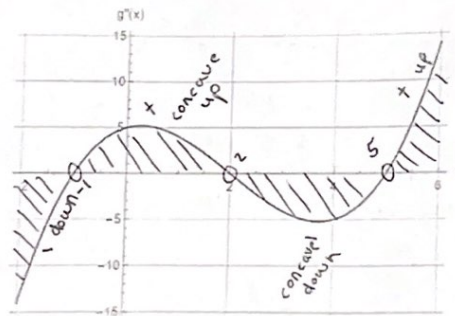
$(x-3)^2 + (y+2)^2 + (z+5)^2 = 38$

QUESTION 18 [2 marks]

(a) Consider the graph of h' :At which value(s) in its domain does the function h have (a) local maximum value(s)?

(1)

$$h(2)$$

(b) Consider the graph of g'' :At which value(s) in its domain does the function g change concavity?

(1)

$$g(-1), g(2), g(5)$$

SECTION C [Full written answers] - 24 marks

QUESTION 19 [4 marks]

- (a) Complete the definition: The function
- f
- decreases on an interval
- I
- if:

(1)

$$f(x_1) > f(x_2) \text{ whenever } x_1 < x_2 \text{ in } I$$

- (b) Given:
- $f(x) = x^3 \ln x$
- ,
- $f'(x) = x^2(3 \ln x + 1)$
- and
- $f''(x) = x(6 \ln x + 5)$
- .

- (1) Find the interval(s) on which the function
- f
- is decreasing.

(2)

$$f' < 0$$

$$x^2(3 \ln(x) + 1) < 0$$

OR:

$$3 \ln(x) + 1 < 0$$

$$\ln(x) < -1/3$$

$$x < e^{-1/3}$$

$$(0, e^{-1/3})$$

$$\textcircled{1} x^2(3 \ln(x) + 1) = 0$$

$$x = 0 \cup 3 \ln(x) + 1 = 0$$

$$\therefore \ln(x) = -1/3$$

$$\therefore x = e^{-1/3}$$

$$\therefore \text{decreases } x \in (0, e^{-1/3})$$

$$\textcircled{2} x > 0$$

	0	$e^{-1/3}$	
x^2	+	+	+
$3 \ln(x) + 1$	DNE	-	+
	DNE	$\ominus$	$\oplus$

- (2) Find the interval(s) on which the function
- f
- is concave upward.

(1)

$$f'' > 0$$

$$x(6 \ln(x) + 5) > 0$$

$$\text{see where } \textcircled{1} f'' = 0 \text{ and } \textcircled{2} f'' \text{ DNE}$$

$$\textcircled{1} x(6 \ln(x) + 5) = 0$$

$$\therefore x = 0 \cup 6 \ln(x) + 5 = 0$$

$$\ln(x) = -5/6$$

$$\therefore x = e^{-5/6}$$

$$\textcircled{2} x > 0$$

	0	$e^{-5/6}$	
x	-	+	+
$6 \ln(x) + 5$	DNE	-	+
	DNE	$\ominus$	$\oplus$

$$\therefore \text{concave up } x \in (e^{-5/6}, \infty)$$

QUESTION 20 [6 marks]

(a) Use logarithmic differentiation to find $\frac{dy}{dx}$ if

(4)

$$y = \frac{3^x \tan(3x)}{\sqrt{x^2 + x^3}} \quad y = \frac{3^x \tan 3x}{\sqrt{x^2 + x^3}}$$

$$\ln(y) = \ln\left(\frac{3^x \tan(3x)}{\sqrt{x^2 + x^3}}\right)$$

$$\ln(y) = \ln(3^x \tan(3x)) - \ln(\sqrt{x^2 + x^3})$$

$$\ln(y) = \ln(3^x) + \ln(\tan(3x)) - \frac{1}{2} \ln(x^2 + x^3)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{3^x \ln(3)}{3^x} + \frac{\sec^2(3x) \cdot 3}{\tan(3x)} - \frac{1}{2} \left(\frac{2x + 3x^2}{x^2 + x^3} \right)$$

$$\frac{1}{y} \frac{dy}{dx} = \ln(3) + 3 \left(\frac{\sec^2(3x)}{\tan(3x)} \right) - \frac{1}{2} \left(\frac{2 + 3x}{x(1+x)} \right)$$

$$\frac{dy}{dx} = \frac{3^x \tan(3x)}{x^2 + x^3} \left(\ln(3) + 3 \left(\frac{\sec^2(3x)}{\tan(3x)} \right) - \frac{1}{2} \left(\frac{2 + 3x}{x(1+x)} \right) \right)$$

(b) Find $\frac{dy}{dx}$ if $y = \tan^{-1}(x+y)$.

(2)

$$\text{let } y = f(x), \quad f'(x) = dy/dx$$

$$f(x) = \tan^{-1}(x + f(x))$$

$$f'(x) = \frac{1 + f'(x)}{1 + (x + f(x))^2}$$

$$f'(x)(1 + (x + f(x))^2) = 1 + f'(x)$$

$$f'(x)(1 + (x + f(x))^2) - f'(x) = 1$$

$$f'(x)(1 + (x + f(x))^2 - 1) = 1$$

$$f'(x) = \frac{1}{(x + f(x))^2}$$

$$\frac{dy}{dx} = \frac{1}{(x + y)^2}$$

QUESTION 21 [4 marks]

Determine the following limits (if they exist):

(a) $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{|x|} \right)$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$\begin{aligned} \lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{x} \right) \\ = 0 \end{aligned}$$

(b) $\lim_{x \rightarrow 0} (1+kx)^{\frac{1}{x}}, k > 0.$

(3)

$$\lim_{x \rightarrow 0} (1+kx)^{\frac{1}{x}}, k > 0 \quad (1^\infty)$$

$$\text{let } y = (1+kx)^{\frac{1}{x}}$$

$$\ln(y) = \ln(1+kx)^{\frac{1}{x}}$$

$$\ln(y) = \frac{1}{x} \ln(1+kx)$$

$$\lim_{x \rightarrow 0} \ln(y) = \lim_{x \rightarrow 0} \frac{\ln(1+kx)}{x} \quad \left(\frac{0}{0} \right)$$

$$\lim_{x \rightarrow 0} \ln(y) \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\frac{k}{1+kx}}{1}$$

$$\lim_{x \rightarrow 0} \ln(y) = k$$

$$\lim_{x \rightarrow 0} y = e^k$$

$$\lim_{x \rightarrow 0} (1+kx)^{\frac{1}{x}} = e^k, \quad k > 0$$

QUESTION 22 [4 marks]

Determine $f(x)$ if f is differentiable, defined on the real line and:

(1) $f'(x) = ax^2 + bx$, with $a, b \in \mathbb{R}$.

(2) $f'(1) = 6$ and $f''(1) = 18$.

(3) $\int_1^2 f(x) dx = 18$

$$f'(1) = 6$$

$$6 = a + b \quad (1)$$

$$f''(x) = 2ax + b$$

$$f''(1) = 18$$

$$18 = 2a + b \quad (2)$$

by solving (1) and (2) we get:

$$a = 12$$

$$b = -6$$

$$\therefore f'(x) = 12x^2 - 6x$$

$$f(x) = \int f'(x) = 4x^3 - 3x^2 + C$$

$$\therefore f(x) = 4x^3 - 3x^2 + C$$

$$18 = \int_1^2 f(x) = x^4 - x^3 + Cx \Big|_1^2$$

$$18 = ((2)^4 - (2)^3 + 2C) - ((1)^4 - (1)^3 + C)$$

$$18 = 16 - 8 + 2C - 1 + 1 - C$$

$$\therefore C = 10$$

$$\therefore f(x) = 4x^3 - 3x^2 + 10$$

QUESTION 23 [3 marks]

Prove that the function f is continuous at a if it is differentiable at a . Explain all your steps. pg 158 in textbook.

Given: $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists

To prove: $\lim_{x \rightarrow a} f(x) = f(a)$

Proof:

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} (f(x) - f(a)) + f(a) \\ &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot (x - a) + f(a) \\ &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x \rightarrow a} (x - a) + f(a) \\ &\quad \underbrace{\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}}_{f'(a) \text{ is a finite number}} \\ &= 0 + f(a) \\ &= f(a) \end{aligned}$$

QUESTION 24 [3 marks]

Assume that $\lim_{h \rightarrow 0} \frac{\cos(3h) - 1}{h} = 0$. Use the definition of a derivative to prove that

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{d}{dx} \sin(3x) = 3 \cos(3x).$$

Explain all your steps.

$$f(x) = \sin(3x)$$

LHS: $\lim_{h \rightarrow 0} \frac{\sin(3(x+h)) - \sin(3x)}{h}$

$$\lim_{h \rightarrow 0} \frac{\sin(3x + 3h) - \sin(3x)}{h} \quad (0/0)$$

$$\lim_{h \rightarrow 0} \frac{(\sin(3x)\cos(3h) + \cos(3x)\sin(3h)) - \sin(3x)}{h}$$

$$\lim_{h \rightarrow 0} \frac{\sin(3x)[\cos(3h) - 1] + \cos(3x)\sin(3h)}{h}$$

End of Section C

$$\lim_{h \rightarrow 0} 0 + \lim_{h \rightarrow 0} \frac{\cos(3x)\sin(3h)}{h}$$

$$\cos(3x) \lim_{h \rightarrow 0} \frac{\sin(3h)}{h} \times \frac{3}{3}$$

$$\begin{aligned} 3\cos(3x) \lim_{h \rightarrow 0} \frac{\sin(3h)}{3h} &= 3\cos(3x)(1) \\ &= 3\cos(3x) = \text{RHS} \end{aligned}$$